

Meysam Hassandoust

Master Student - Sharif University of Technology

@ Meysamryn@gmail.com +98 9388794783 Tehran, Iran meysam-hassandoust.github.io



Education

Sharif University of Technology

M.Sc. in Physics

2024 – present Tehran, Iran

- GPA: 4/4
- Ranking: 1st

Shahid Beheshti University

B.Sc. in Physics

2019 – 2024 Tehran, Iran

- GPA: 3.5/4 (Physics)
- Ranking: 1st

Selected Courses

- Advanced Quantum Mechanics
- Advanced Electrodynamics
- Advanced Statistical Mechanics
- Quantum Field Theory 0
- Quantum Fields Theory 1
- String Theory (Lecture Notes)
- Superstring Theory (Lecture Notes)
- M-Theory (Lecture Notes)
- Algebraic Quantum Field Theory (Lecture Notes)

Research Experience

Review of Algebraic Quantum Field Theory

Sharif University of Technology

2025-2026 Tehran, Iran

- Supervisor: Dr. Amin Faraji Astaneh
- We gave an overview of entanglement, von Neumann algebra, and the emergence of spacetime.

The Imaginary Phase Problem in the Gravitational Path Integral

Sharif University of Technology

2024-2026 Tehran, Iran

- Supervisor: Dr. Amin Faraji Astaneh
- In the gravitational path integral, the saddle point approximation introduces Polchinski's phase, which confuses state counting. In de Sitter gravity, an observer's perspective yields a von Neumann algebra type II, hinting at a connection between Polchinski's phase and gravitational state counting.

Areas of Interest

- String Theory,
- Quantum Gravity,
- Mathematical Physics.

Skills

- Python, C/C++, HTML/CSS, Mathematica, $\LaTeX$
- Problem Solving, Effective Communication, Teamworking

Honors

The University Students National Physics Olympiad

IPhO

2024 Iran
Ranked 14th (Silver Medal)

The University Students National Physics Olympiad

IPhO

2023 Iran
Ranked 8th (Silver Medal)

The National Master Studies Entrance Exam

National Organization of Educational Testing

2023 Iran
Ranked 14th and 18th in two consecutive years

Gold Medal World Mathematics Team Competition

WMTC

2013 Iran

A generalization of the Selberg zeta function for flat space cosmologies

Shahid Beheshti University

📅 2024-present 📍 Tehran, Iran

- Supervisor: Dr. Reza Fareghbal
- We construct a generalized Selberg zeta function in flat space cosmologies(FSC's) and obtain a 1-loop partition function in the scalar field in FSC's.

Group theory and irreducible representations of the Poincare group

Shahid Beheshti University

📅 2023 – 2024 📍 Tehran, Iran

- Bachelor thesis
- Supervisor: Dr. Siamak Sadat Gousheh
- We started from the basics of group theory and obtained their irreducible representation for Poincaré groups.

Supersymmetry in Quantum Mechanics by Generalized Uncertainty Principle

Shahid Beheshti University

📅 2023 – 2024 📍 Tehran, Iran

- Supervisor: Dr. Siamak Sadat Gousheh
- We investigated supersymmetry in quantum mechanics with the generalized uncertainty principle.

Supersymmetry in Quantum Mechanics for Dirac equation in FRW space-time

Shahid Beheshti University

📅 2022 – 2023 📍 Tehran, Iran

- Supervisor: Dr. Siamak Sadat Gousheh
- We constructed supersymmetry quantum mechanics in curved space-time and investigated what conditions can preserve this symmetry in FRW space-time.

Reference

- Dr. Amin Faraji Astaneh
- Dr. Siamak Sadat Gousheh
- Dr. Seyed Ali Hosseiny Esfidavajani
- Dr. Ali Sadeghi
- Dr. Reza Jafari
- Dr. Amir Kargaran

Teaching Assistant

Electromagnetics III

Sharif University of Technology

📅 Spring 2026

📍 Tehran, Iran

Physics Laboratory I

Sharif University of Technology

📅 Fall 2024

📍 Tehran, Iran

Quantum Mechanics I

Shahid Beheshti University

📅 Fall 2022

📍 Tehran, Iran

General Physics I

Shahid Beheshti University

📅 Fall 2022

📍 Tehran, Iran

Electromagnetism II

Shahid Beheshti University

📅 Spring 2021

📍 Tehran, Iran

Mathematical Physics II

Shahid Beheshti University

📅 Spring 2021

📍 Tehran, Iran

Analytical Mechanics II

Shahid Beheshti University

📅 Spring 2021

📍 Tehran, Iran

Mathematical Physics I

Shahid Beheshti University

📅 Fall 2020

📍 Tehran, Iran

General Physics III

Shahid Beheshti University

📅 Spring 2019

📍 Tehran, Iran

Languages

Persian ●●●●●

English ●●●●●